



The Platform Society

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CHAPTER

6 Education

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Abstract

This chapter investigates how platformization is affecting the idea of education as a common good on both sides of the Atlantic. The growth of online educational platforms has been explosive, in both primary and higher education. Most of these educational platforms are corporately owned, propelled by algorithmic architectures and business models. They have quickly gained millions of users and are altering learning processes and teaching practices; they boost the distribution of online course material, hence impacting curriculums; they influence the administration of schools and universities; and, as some argue, they change the governance of (public) education as a whole. The chapter explores how, powered by the Big Five, these educational platforms are pushing a new concept of learning that questions values that are fundamental to publicly funded education: *Bildung*, a knowledge-based curriculum, autonomy for teachers, collective affordability, and education as a vehicle for socioeconomic equality.

Keywords: education, Bildung, learning, MOOC, Coursera, AltSchool

Subject: Economics, Digital Lifestyle, Media Studies

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Introduction

After dealing with platformization of news, urban transport, and health in the previous chapters, we now turn to a sector that has traditionally been part of the public domain. Public education has long been one of the most precious common goods and the backbone of Western democracies—a backbone that was scaffolded by independent institutions like schools and colleges, which employed teachers who autonomously decided upon a curriculum for their students. Those curriculums were rooted in public values, such as *Bildung*—the ideal to teach children to become not just skilled workers but **knowledgeable citizens**—and equality—each child getting access to affordable education. Although there are considerable differences between the United States and western Europe in terms of how and how much public school systems get funded by their respective states, their underlying ideals tend to be similar.

In this chapter, we investigate how platformization is profoundly affecting the very idea of education as a common good on both sides of the Atlantic. Over the past few years, the growth of online educational platforms has been equally explosive as the growth of platforms in the health and fitness field. Most of these educational platforms are corporately owned, and propelled by algorithmic architectures and business models. They have quickly gained millions of users and are altering ↪ learning processes as well as teaching practices; they boost the distribution of online course material, hence impacting curriculums; they influence the administration of schools and universities; and, as some argue, they change the governance of (public) education as a whole (Williamson 2016a, 123). Powered by the Big Five tech companies, platforms are pushing a new concept of learning that uproots or bypasses the values that are fundamental to publicly funded education: *Bildung*, a knowledge-based curriculum, autonomy for teachers, collective affordability, and education as a vehicle for socioeconomic equality.

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In the next section, we will explain how sectoral educational platforms are developing as part of the larger connective infrastructure, dominated by the Big Five platform operators and carrying the ideological imprint of the ecosystem's architecture. More recently, **global tech firms have also started to seriously invest in the education sector**, either through distributing hardware and software to schools or by means of their charity foundations targeting educational systems. Therefore, we analyze *how* the platform mechanisms—datafication, selection, and commodification—influence traditional principles of learning and teaching. To see how these mechanisms work in the practice of school systems, we will look at two different examples concerning two types: primary schools and higher education.

AltSchool is an educational startup targeting K–12 schooling in the United States; the initiative is backed by Silicon Valley investors and has been touted the “Uber for primary education” because of its potential to scale. The technological architecture of the platform directly informs its pedagogical philosophy of personalized data tracking; learning processes are translated into data processes and turned into tracking systems that continuously relate individual progress to standardized performance. Next, we will turn to higher education and explore how **massive open online courses (MOOCs)** have affected global online markets and the organization of (public) universities. American educational platform Coursera deploys the principles of datafication and personalization, while its business model explicitly fosters global expansion. MOOCs are promoted as an efficient and effective form of higher education—allegedly cheaper than courses and curriculums offered by brick-and-mortar universities. Zooming in on the principles of commodification, we will particularly pay attention to how this model affects public values.

When talking about public values in the sector of education, we are tempted to concentrate exclusively on parents' and students' concerns about privacy, security, and accuracy of the data processed by apps. Educational data, along with health data and financial data, are among the most valuable currencies in the ecosystem, so safeguarding these values is extremely important. However, as stated in the opening paragraph, public values in this sector have a much wider reach. Platformization is likely to redefine education as a common good as it gets caught between two ↪ ideological sets of values: *Bildung* vis-à-vis skills, education versus learnification, teachers' **autonomy** versus automated data analytics, and public institutions versus corporate platforms. In the last section, we will discuss how datafication and personalization can be leveraged toward *enhancing* the common good, if organized as “open” initiatives and supported by public institutions, governments, and civil society actors.

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Educational Platforms as Part of a Connective Ecosystem

Educational platforms cannot be seen apart from the larger ecosystem of connective media on which they are built and through which they thrive. As we explained in chapter 1, Facebook, Alphabet-Google, Apple, Amazon, and Microsoft are pivotal gatekeepers in the data flows that fuel the online educational sector which is turning into a booming global market. So it should come as no surprise that three of the Big Five tech companies have invested in the educational sector from the very start of their business. Microsoft has been active in educational software since the 1980s, most notably through its office software and classroom software (i.e., Blackboard). Apple has built its imperium partly on the popularity of its hardware devices, such as laptops, desktops, and tablets in classrooms, drawing children and young adults into its iOS-based universe. The introduction of specially designed software for its popular handheld devices is Apple's latest example to enter the market for preschoolers. Google, from the very beginning of the company in 1999, has held a significant presence in students' lives through offering free services to universities, such as Scholar, LibraryLink, Gmail-and-Docs packages for students, and many more. In terms of hardware, the use of cheap Google Chromebook laptops has become widespread in K–12 schools across the globe.¹

More recently, high-tech companies have moved on from hardware and software to the next generation of data-driven, platform-based tool sets—tools that further penetrate the core business of teaching: content production and distribution, student performance tracking, class communication and administrative organization. For the past five years, the market for online educational services has flourished as schools have been rushing to implement sophisticated portals that allow students to see course assignments, let teachers communicate with students, offer digital courseware, and permit parents to assess their children's progress.² Intricate data-driven products are designed to adapt to the abilities and pace of each individual child, holding out the promise of improved academic achievement. Apple introduced the comprehensive Classroom App, a toolset for managing and monitoring its iPad products in the classroom. Microsoft presented MS Education, an online toolkit ↵ for schools that “empowers students to develop modern skills, including communication, collaboration, problem solving and global awareness” (Microsoft Education 2017). Google created its own market niche of combined hardware and software when it preinstalled Google Apps for Education on its Chromebooks. In 2016, Amazon—a relative newcomer to the online education market—launched Inspire, a portal where teachers can upload, share, and review free digital learning material, not only books from the Shakespeare Library but also interactive tools from museums and open source materials, such as College Scorecard.³ By launching Inspire, Amazon joined its high-tech rivals in their push to expand the use of digital technology in public and private schools, both in the United States and beyond. A number of tech entrepreneurs, most notably Facebook's Marc Zuckerberg, have recently invested their fortune in charity or nonprofit funds that promote the platformization of education and expressly the teaching of computer skills, math, and coding.⁴

In the online educational market, the Big Five companies are competing not just with each other but also with established education companies such as Pearson and Knewton, which have vested interests in holding on to their market base (Williamson 2016a). Like legacy news organizations, these companies are confronted with a new type of rival that has a huge lead in cross-sectoral data processing and platform distribution; as said before, they control the ecosystem not only through their infrastructural services (browsers, cloud services, search engines, etc.) that already have a strong presence in educational contexts but also via their partnerships with, or acquisitions of, promising ed-tech start-ups. There are still a fair number of successful independent developers and start-ups in the higher education segment.⁵ However, few initiatives are able to scale their products without the support of at least one of the five big players. Google and Facebook have particularly capitalized on online educational services that aim to transform both the content and context of learning.

Sketching a rough outline of how the educational sector is embedded in the larger ecosystem of infrastructural platforms is helpful to understand how their integration with sectoral connective platforms helps sustain the

dominant position of the Big Five platforms through path dependency. But we need to dig deeper into this dynamic to understand how they influence the fabric of social structures and institutions. Governing the ecosystem of platforms are the three mechanisms—datafication, selection, and commodification—that drive the reorganization of societal sectors.

p. 121 Datafication stands out as the single most important factor in the transformation of online education, changing school curriculums from kindergarten to university. Large quantities of data are generated and collected automatically through the devices and services that students, teachers, parents, and school administrators use. Data sets range “from the often ad hoc ‘in-house’ monitoring of students and teachers to the systematic ‘public’ collection of data at local, state and federal levels” (Selwyn 2015, 66). As educational assessment is increasingly quantified, student performance gets governed by numbers (Robert-Holmes 2015). Quantitative learning analytics, grounded in large numbers of students’ online behavioral data, involve *real-time data tracking* as well as *predictive analytics* (Tempelaar, Rienties, and Giesbers 2015). Data tracking can be used to register fine-grained information about the time a student needs to solve a problem, to record the cognitive stages in problem-solving, to measure the amount of instruction needed, or to trace student interaction (Koedinger, McLaughlin, and Stamper 2014). Such computations can help predict an individual’s chances for academic success based on average scores and a number of variables. These emerging digital policy instruments outsource the assessment of didactic and pedagogical variables from teachers and classrooms to data scientists and algorithms. **The general idea of data-driven platform services is to allow for modification and corrective action of learning activities; they provide the as yet unrealized potential to radically transform learning processes and pedagogies** (Maull, Godsiff, and Mulligan 2014).

Datafication is a precondition for personalization—a subcategory of what we identified in chapter 2 as the mechanism of selection. Personalized learning means that the online system adapts to each student’s learning needs and abilities, to optimize individual performance and boost motivation. Although personalized learning is nothing new, in the context of user-data collection and predictive analytics it means that continuous individual monitoring and tailored didactics become integral to the pedagogical model. The ability to automatically track individuals obviously includes competence scores and performance levels but may also pertain to stress levels, speed variations, and heart rates. Personalized data allegedly provide unprecedented insights into how individual students learn and what kind of tutoring they need. If the data show that one student is better at solving verbal math problems than abstract equations, software developers can adapt the assignments and implement personalized variations. Aggregated data about learning behavior provide the input for individual “adaptive learning” schemes. Personalizing the learning experience is often embraced as the quintessential motivation booster; it lends its focus from intelligent business analytics and customer satisfaction measurements.

p. 122 Datafication and personalization can hardly be analyzed separately from the mechanism of *commodification*: the monetization of content and data flows via their platforms’ business models and governance structures. Commodification in the context of educational platforms often involves the processing of learning data by large data hubs that render big data streams monetizable and potentially profitable.⁶ Massive information transfer to “calculation centers” and data hubs tallies with the concentration of power in a few large data companies which serve commercial rather than pedagogical interests (Lawson, Sanders, and Smith 2015). As illustrated by some of the Big Five sectoral connectors, for instance, Google Apps for Education and Amazon Inspire, their interface feature design is strikingly similar to those inscribed in their (commercial) infrastructural services, affording a smooth exchange of data flows (Friesen and Lowe 2012; Lindh and Nolin 2016). Data flows generated by students and teachers have high economic value because they provide precious insights into individual vis-à-vis group preferences. Children and young adults are extremely valuable user populations, not least because they form a coveted but vulnerable target audience for advertisers and service providers—an extra reason for policymakers to be vigilant with regard to student privacy and security.

In principle, commodified online services based on the mechanisms of datafication and personalization have great potential to advance education, making it more efficient, accessible, and affordable; but there is no evidence yet that they will significantly improve learning and academic outcomes for the majority of students over the long term (Beetham and Sharpe 2013). The effects of data-driven, personalized education are as of yet scarcely researched. Proponents of quantitative learning analytics claim they are a significant quality booster for student learning (Ebner and Schön 2013; Reamer et al. 2015). Real-time data about individual learning processes help instructors monitor students' progress and allow for corrective feedback. As some advocates argue, continuous measuring of performance levels may enhance the quality of instruction (Mauß, Godsiff, and Mulligan 2014). Others hail the datafication of educational processes, emphasizing how personalized data analytics inform learning design and facilitate specific pedagogical action (Lockyer, Heathcote, and Dawson 2013).

Detractors from across the world have pointed out how a growing emphasis on datafication through online services can be ineffective in terms of boosting student performance or academic levels; a gloomier prediction is that they intensify managerialism and the surveillance of students and teachers (Selwyn 2016; Thompson 2016). In a 2016 British government report, *From Bricks to Clicks*, a number of critics draw attention to the ethical and legal challenges involved in data collection and interpretation, including privacy, security, and control over one's own data (Higher Education Commission 2016). These important public values have also been at the forefront of researchers' scrutiny of data-intensive learning analytics (Siemens 2013; Slade and Prinsloo 2013). However, the criticism carries beyond privacy and security. Some educators critical of the UK-government report have questioned the very assumptions of data-driven, personalized education as pillars of trust and public confidence (Butcher 2016). **They argue that the government's uncritical endorsement of platformization ignores basic principles such as academic autonomy and independence—principles on which institutions for public education are founded.** If dashboards for monitoring students' individual progress become the prime instruments for teachers, instead of classroom activities, this could cause a paradigm shift in education (Thompson and Cook 2016).

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In short, while acknowledging the positive uses of datafication and personalization in teaching, critics draw attention to the fact that none of the presumed benefits cited by platforms have been proven empirically. The changing role of teachers from classroom directors to dashboard controllers, mediated by numbers and analytical instruments, is a major issue; professionals may feel that the core of educational activities—assessment and personalized attention—gets outsourced to algorithms and engineers. Let us now look more closely at two examples to help clarify what public values are at stake in the platformization of education.

AltSchool and the Transformation of K–12 Education

AltSchool is a Californian initiative founded in 2013 by Google's former head of personalization Max Ventilla; the venture has drawn much attention, if not for its educational philosophy then certainly for the names of its financial backers, including Mark Zuckerberg's education-focused nonprofit foundation and the Founder's Fund of venture capitalist Peter Thiel. AltSchool is a hybrid physical–virtual initiative. In 2017, it operated six physical micro-schools—one-classroom settings that house teachers and pupils but no administrators or secretaries; administrative and managerial tasks are handled by the platform's headquarters in San Francisco.⁷ Its “virtual” products allegedly have great scaling potential, promising to benefit public school systems in the future. The investors' objective is to help reinvent American education, “first, by innovating in its micro-schools; next, by providing software to educators who want to start up their own schools; and, finally, by offering its software for use in public schools across the nation, a goal that the company hopes to achieve in three to five years” (Mead 2016, 6).

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More than just a classroom, AltSchool is an experimental lab where teachers and students are minutely observed; each pupil has an iPad or Chromebook, and every activity is automatically recorded and analyzed. A large staff of over one hundred technicians and data scientists develops customized educational software and tweaks learning analytics to optimize each part of the learning process. The tools developed by this team are remarkably similar to those deployed by general networking platforms: a recommendation algorithm not unlike Netflix or Amazon that takes into account everything a child has already learned, data analytics that eliminate the need for regular testing, and reputation mechanisms that rate the personal input ↵ of children while measuring their relative scores.⁸ Teachers supervise the learning process; group instruction is replaced by independent projects. Instructed through a weekly “playlist,” each pupil engages in individual or group activities that are tailored toward his or her unique capabilities.

In this high-tech educational setting, student performance is monitored through tracking systems that minutely evaluate each individual child’s skills. This means for some pupils that math gets prioritized over reading skills and for other children the other way around (Lapowsky 2015). Informed by data, teachers keep scoring “cards” of each child’s progress—academically, emotionally, and socially—and use them to make weekly personalized learning plans. The process of datafication supposedly renders education more efficient by cutting out ineffective tasks for each individual child. In terms of educational philosophy, AltSchool favors technology over teachers, online task-oriented learning takes over classroom instruction, and predictive analytics replace teachers’ professional judgments.⁹ In addition, individualized learning schemes take priority over a common curriculum or educational canon. To be sure, a personalized approach of students’ learning behavior and a differential treatment of educational content are not exclusive to platform innovations like AltSchool. The very principles of datafication and personalization can be helpful aids to teachers if applied in a local setting; teachers can perform detailed assessments of student progress without having to rely on globally scaled connective tools developed by tech companies.

Besides the disputed effects of connective tools, some have pointed at the way in which platforms like AltSchool advance a pedagogical perspective also known as “learnification.” Many data-driven, personalized education initiatives focus on *learning* rather than on education and on *computable processes* rather than on **social interaction between teachers and students**. The (social) activity of learning is broken into quantifiable cognitive and pedagogical units, such as instruction, short quizzes, assignments, deliberation with other students, and tests. The learnification model is predicated on the real-time, short-term process of learning, while education involves a simultaneous nourishing of intellectual, social, technical, and cognitive skills and involves a longer trajectory over a period of years. The “learnification” paradigm, according to some social scientists and philosophers, sprouts from the idea that learning can be managed, monitored, controlled, and ultimately modified in each student’s personal mind (Attick 2013; Biesta 2012; Mead 2014).

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Datafication coupled with personalization appears to be a double-edged sword. Customized assignments based on quantitative measurements may help improve each individual’s learning process and outcomes. Still, personalization algorithms are based on *inferences* of users’ needs or interests. So a by-product of personalized learning algorithms may be that they “filter out what is not designated as being of ↵ interest to users and rather presents to them only what fits the system’s belief of what their interests are” (Ashman et al. 2014, 824). In other words, personalized assignments may be the educational equivalent of filter bubbles in the news sector. One could argue that AltSchool delegates professional decision-making to data-driven algorithmic systems that have the capacity to circularly define, predict, and govern learners’ progress (Williamson 2015). In her ethnographic observation of AltSchool’s daily practice in the *New Yorker*, Mead (2016) concludes that data analytics change the role of an educator to “someone who is more of a data-enabled detective” (9).

Besides criticizing the learnification paradigm, researchers have raised various other concerns with regard to platformization, such as surveillance (Siemens 2013). Student and teacher performance are increasingly monitored through dashboards; constant assessments are based on behavioral tracking mechanisms and automated classroom surveying techniques introduced in many elementary schools throughout the United

States that have triggered serious controversy (Singer 2015; Rich 2015). Students enrolled in AltSchool and similar platforms are prone to constant monitoring, to the extent that they become (unwitting) participants in continuous lab experiments. Over the past few years, online education has shown a conclusive shift toward continuous behavioral experiments (Ebben and Murphy 2014; Singer 2017a). Using students for research experiments is nothing new. However, the *continuous* tracking of young children triggers important ethical questions; issues of privacy and repurposing of data raised anxieties with parents and triggered inquiries from regulators (Selwyn 2015).

Other questions scarcely addressed by researchers are the ethical issues involved in the ownership of (meta)data and privacy protection, which can be quite thorny. In datafied school systems, students' performance is assessed not just in terms of grades or learning outcomes but also in terms of learning curves and social adaptability. Datafication of the learning process—minutely monitoring the interactive and cognitive behavior of pupils—yields an abundance of data beyond mere academic test results. How fast do students answer a quiz question? How solitary or collegial are they when it comes to problem-solving? Behavioral data are a sort of by-product of continuous monitoring, and students are barely aware of these data being accumulated, interpreted, and repurposed (Ashman et al. 2014). In fact, dashboards may produce behavioral information that is far more interesting than a straightforward grade point average. It is not at all difficult to imagine how future employers might request full performance records from kindergarten up to college to assess an employee's intellectual and social adaptability. The significance of privacy in an educational context cannot be overstated; as Helen Nissenbaum (2010) has pointed out, privacy is a contextual value.

p. 126 Students need a comfortable setting in which they ↵ can make mistakes or take time to discover new ideas and acquire skills. If data about learning behavior get commercialized or traded in any way, students' comfort zone is seriously infringed upon.

AltSchool's use of platform dynamics is typical for many online initiatives currently being developed and implemented. And AltSchool is not the only pet project of Silicon Valley investors that promises to scale its technologies to empower all children in public schools. In 2015, Facebook initiated a partnership with California Summit Public Schools to engage with its K–12 educational program to pioneer a datafied personalized learning model similar to Altschool's.¹⁰ Despite pledges to comply with privacy regulations and to protect student data, the interest of tech companies in the scalability of educational online systems promoting datafication and personalization can hardly be understood apart from the mechanism of commodification; as Ben Williamson (2016b) argues, these “start-up schools” epitomize “the next step in corporate education reform by venture philanthropy” (4). The platformization of primary education—including its potential benefits and possible drawbacks—should be assessed in light of the larger question concerning the privatization of public education. Particularly in Europe, where publicly funded schools are the norm, the effects of corporate platforms and their mechanisms should be subject to regulatory scrutiny and public debate. In the next section, we will try to understand the intricate dynamics of educational platforms as well as their impact on public institutions, shifting our focus to the field of higher education.

Coursera and the Impact of MOOCs on Higher Education

The platformization of higher education has manifested itself in a variety of ways, but one of the earliest forms was the quick rise of MOOCs. These popular vehicles for online learning were initially promoted as future alternatives to, or even replacements of, college education. From their very onset in 2012, US-based platforms such as Coursera, edX, and Udacity positioned their platforms as global ventures, some of them based on for-profit models and others organized as nonprofit collaborations between universities and high-tech companies.¹¹ MOOCs did not come out of the blue; they fit a long tradition of distance learning and later e-learning in the United States and on the European continent. Between 2013 and 2015, the global interest for MOOCs surged as they managed to each occupy a specific market niche.¹² Even if the popularity of MOOCs may seem to have worn off a bit, their impact on higher education as a global online market still reverberates across the United States and Europe.

p. 127 For this section, we will zoom in on Coursera, a for-profit platform that started in 2012 out of Stanford and has offered thousands of online courses to students worldwide.¹³ Coursera deploys many of the same datafication and selection mechanisms as previously identified; the platform equally subscribes to the principles of learnification and vents out personalized learning strategies driven by predictive analytics. In an earlier article, we described extensively how the mechanisms of datafication and personalization are driving the development of online MOOCs (Van Dijck and Poell 2015). For the purpose of this chapter, we will concentrate on the mechanism of commodification, which is driving Coursera's business model. Similar to Facebook, Google, and other major platforms, Coursera offers content that is preferably free of charge to end users; students can enroll and watch videotaped lectures, engage in online exercises, and take tests. Coursera's goal is to connect educational content to a global mass audience. User acquisition is critical to achieve network effects, so MOOCs have a vested interest in drawing massive numbers of users in order to obtain big data sets and many student profiles. The more (meta)data they collect, the more they learn about their users (Koedinger, McLaughlin, and Stamper 2014; Clow 2013). But how does Coursera derive revenue from its activities?

MOOCs may be free to end users, but they do not come cheap.¹⁴ Over the past years, Coursera has experimented with various business models and is still fine-tuning its monetizing schemes. First of all, the platform's revenue sources are predicated on the efforts of those universities and colleges it collaborates with. It is interesting to contrast Coursera's business model with that of legacy institutions—private and public universities—which offer a “bundled” educational experience; this experience comes complete with lecture halls, facilities, libraries, and labor-intensive courses that include certification, advising, tutoring, and testing (Boullier 2012; Decker 2014). For this all-inclusive package, students pay tuition and fees; in the case of public education, taxpayers fund the institutions to make higher education affordable and accessible to more citizens.¹⁵ The conventional university business model reflects the ideology of higher education as a curriculum-based, comprehensive experience that focuses on *Bildung*; rather than churning out skilled workers, **public education promotes the education of knowledgeable and critical citizens—a condition for healthy democracies**. Most institutions of higher education are still grounded in the principle of curriculum-based diploma or degree programs, even if many universities are also offering single courses through extension programs or special programs tailored to specific groups.

p. 128 For MOOC platforms, the core unit of education is not *Bildung* but *learning*; the core unit of trade is not a *curriculum* but a *course*—a single unit that can be “unbundled” and “rebundled” into an online “product” that is offered outside of a local college ↪ context to an audience spread across the globe. Coursera's model parallels the way in which connectors like Facebook and Google promote the “unbundling” of news content, as described in Chapter 3: news articles and advertisements no longer come as a package (newspaper) but are redistributed via Facebook's Instant Articles News Feed, and Google News. Like in news production, Coursera positions itself as a connector in a multisided market, connecting content, students, and third parties. But who pays for the online product? Content is produced and paid for by universities consorting with Coursera;

universities commonly remunerate lecturers for the development of course content as well as instructors who process student feedback and questions. Universities and lecturers are thus complementors to these online connectors. In collaborating with platforms, universities can decide to offer some online courses for free while making students pay for other content. If universities ask user fees, they have to share revenue with the platform.¹⁶ Universities, in other words, invest heavily in Coursera's business model, paying for content and teaching efforts, while the platform gets compensated for its "connective services."¹⁷ Just as Uber has no cars and employs no drivers, Coursera employs no teachers and owns no school buildings.

Although the "free" model reigns in the world of MOOCs, a premium variant expands the free option by offering extra paid-for services such as so-called signature tracks, where students pay for certificates of completion, proctored exams, and identity verification. Proctored exams and verified certificates are gradually developed into—potentially profitable—units that can be marketed globally. These micro-degrees or nano-degrees are becoming very popular, particularly in markets where they target professionals who want to learn specific skills (Kalman 2014). What we are currently witnessing in the United States is education platforms teaming up with high-tech companies to train a skilled workforce: Coursera, in collaboration with Google, Instagram, and others, offer so-called capstone projects—online internships—where students take on assignments from companies to develop tech skills as well as marketing skills. Besides the labor that goes into these projects, the data gathered from the students' learning process are highly relevant to companies looking for future employees (Lardinois 2015).

p. 129 Arguably the most profitable business model for online educational content is to collect and trade user data. Coursera's value proposition runs parallel to those of most social media platforms: money is made from leveraging data value to interested third parties. Every user signing up with the platform checks a box, allowing Coursera to collect all user data.¹⁸ It is unclear how Coursera is currently monetizing this wealth of data. Inserting advertising in online content is an option that Coursera has not utilized yet, but it may well be a lucrative value proposition in the future; already, Coursera is fully absorbed in the platform ecosystem, where Facebook and Google dominate online advertising (Kalman 2014). As we saw in the example of AltSchool, the accumulation of behavioral learning data culled from participants can be sold to businesses competing for global talent. The value of learning data is enormous; combined with other data, such as social media profiles or health and fitness data, personalized information is a gold mine not only to prospective employers but also to insurance companies or consumer firms. The repurposing of student data in the US context is largely a gray legal area; as Young (2015) points out, privacy law is outdated and insufficient when it comes to platforms like Coursera or other MOOCs.

If we look at the mechanisms underpinning the development of MOOCs, we can hardly deny their disruptive force in the global landscape of higher education. Many regard the tendency toward offering personalized, datafied, and customized online learning materials as a positive development. The idea of offering courses which can be taken on demand, apart from an institutional context, is a very attractive scheme in a world where lifelong learning skills are increasingly important and where a decent college education, especially in the United States, has become a privilege that many aspire to but few can afford. The higher the cost of the conventional college package deal, the more attractive MOOCs' "unbundling" proposition becomes. For individuals unable to enroll in traditional colleges the alternative of buying the "derivative" of a much coveted college degree may be very attractive.

As much as MOOCs may be complementary to current university programs and benefit *individual* students, they may undermine some of the core values of *collective* public systems in the long run. While few people believe MOOC platforms pose a threat to brick-and-mortar educational institutions, they are likely to have a strong impact on education as part of the public sector—particularly in Europe where public education is still considered an important contribution to the common good. Although it is impossible to predict how deeply the dynamics of platformization will penetrate the traditional structures of higher education, two potential implications follow from the analysis above. First, public schools and colleges have been great equalizers,

proven engines of upward mobility due to their affordability and equal-access policies. Second, while schools and colleges were granted autonomy in defining their curriculum, guarding the standards of education was always a public responsibility, overseen and executed by independent agencies. Platformization tends to disrupt these public values of equal access, professional autonomy, and accountability.

p. 130 To start with the former, MOOCs are often presented as high-tech fixes to a crumbling public institution—an institution that fails disadvantaged and poor students and has become less accessible to poor students due to rising costs.¹⁹ Coursera deploys compelling rhetoric to promote the platform’s scaling capacity to make higher education available to less advantaged students all over the world, particularly those from developing countries. So far, there is little evidence to prove that MOOCs improve the accessibility or affordability of education.²⁰ Indeed, MOOCs cause “new relationships in the geometries between the learner, the developer of the content, the delivering and the assessment” of education (Jacovella 2013, 1300). But one way or another, that “new geometry” is deeply reliant on quality content provided by quality teachers that has to be paid for either by student fees or by collective taxes. The much broader argument that the money needed to fix public schools should be spent on private charters and high-tech fixes has become an increasingly popular tenet, especially in the United States, where it is leading to a dismantling of the public system.

Another embattled public value at stake here is the professional autonomy of teachers to design their own content in light of a globally operating, highly centralized and standardized distribution platform. Indeed, Coursera’s scaling capacity is driven by a commercial business model, but even in the case of nonprofit MOOCs, the urge to operate on a global scale seriously affects the autonomy of teachers to design courses according to their own pedagogical principles. For instance, edX is the largest nonprofit MOOC venture.²¹ Unlike Coursera, edX initially resisted the pressure of venture capitalists to scale and make money quickly, resulting in a focus on experimentation and collaboration with teachers toward building a sustainable online environment. However, as Kelkar (2017) has demonstrated in her ethnographic study of edX practices, over the course of several years there has been mounting pressure from “edX architects” to improve the platform’s scalability and global standardization. Meanwhile, instructors and participating college administrations saw a gradual decline in their “ability to set the agenda and control the direction of the software” (Kelkar 2017, 14). EdX system architects admit they struggle with the tension between serving their paying customers and the demands of the wider community (e.g., open source community, researchers).

p. 131 Finally, the third contested public value is accountability: the role of independent calibration and accreditation institutions in guarding the value of educational credits and diplomas. Regulation of quality standards in education is still largely handled through national agencies and, especially in Europe, through government institutions that are subject to democratic control. Platformization in a connective world may lead to the privatization of accreditation and certification systems. Coursera, for one thing, is keen on developing worldwide standards for credits and exams; much like Facebook wants to offer a standard for identity verification, Coursera has invested in acquiring a gatekeeping position in the field of accreditation and validation. Since Coursera operates on a global scale while many accreditation systems are still regulated by national or supranational (European) agencies, there is a potentially lucrative market. Uber and Airbnb have proven they can disrupt entire sectors by fighting regulatory conflicts at the national or local level; in organizing its own global accreditation system, Coursera may simply bypass regulation and thus undermine the public sector’s governance.

Education as a Common Good: Blended Learning and Open Source

The transition toward a platform society disrupts traditional pedagogies of learning and teaching. Using AltSchool and Coursera as examples of how the mechanisms of datafication, personalization, and commodification work, it is important to realize how platformization affects the notion of education in the service of the common good. The two examples bring to the fore an intrinsic clash between two conflicting ideological views on the value of education: on one end of the spectrum, education is a private investment in youngsters, to teach them skills and enhance their employability; on the other end, education is a public investment meant to **foster knowledgeable and critical citizens and enable them to participate in a democracy**. In practice, the opposite views are obviously never this clear cut. Public values, as we argued in the first chapter, are the stakes in a negotiation over what the platform society should look like. So far, we have looked mainly at how the mechanisms of datafication and commodification imprint certain values in the architecture of the platform ecosystem. But what have other (nonmarket) actors in the platform society—governments, public institutions, and civil society actors—contributed to this negotiation? And what are their preferred tools and principles when it comes to procuring education as a common good, particularly in Europe where education is firmly entrenched as a public sector?

To start with the latter, nonprofits and governments primarily advocate two preferred principles: blended learning and open source. Digital and online learning tools can be extremely valuable additions to the arsenals of teachers and researchers, if carefully mixed with proven pedagogical methods that are implemented locally. So-called blended learning environments may enhance the learning process, not only for already advantaged students but also for those who are less gifted and need more training (Gikas and Grant 2013; Rennie and Morrison 2013). The focus on blended learning has caused a shift from *massive* open online courses (MOOCs) to *small* private online classes, leaving more autonomy to teachers and emphasizing the importance of human interaction between teacher and students. To be sure, many of these experiments are still in full swing, and there is no empirical evidence yet that “flipping the classroom” enhances the broader goals of (public) education.

p. 132 Most universities and several nonprofit educational platforms are advocates of open source software; edX, for instance, has opened up not just its courseware but also the source *codes* of its tools to everyone who wants to use them. As we saw in chapter 5 on health apps, nongovernmental organizations, governments, and public institutions have turned “open source,” “open data,” and “open knowledge” into the mantras of the platform society. Research organizations, hospitals, and universities want to create a sustainable online infrastructure, in terms both of content distribution and of data management systems that support the public values of findability, accessibility, interoperability, and reusability.²² But beyond those immediate operational values, it is mainly up to governments to guard the societal values of education: to promote the fair use of data while protecting students’ privacy, to enhance education rather than just learning, to maintain independent quality control, and to keep education affordable.

Governments have also taken up their responsibility to defend education for the common good in the platform society by assuming an active role as platform developers, trying to streamline organizational processes while stimulating innovation (Janssen and Estevez 2013). Over the past few years, there has been a growing number of public initiatives in the educational domain, both by public school systems and by governments that eagerly invest in so-called open platforms, some aiming at primary schooling but most targeting higher education.²³ At the national level, Spain, Italy, and Poland have launched platforms for online courses in direct response to American MOOCs. In 2013, a pan-European public alternative called OpenUpEd was started in the United Kingdom, in an attempt to counterbalance US corporate dominance in this area.²⁴ Some of these platforms have had moderate success, befitting the modest investments from public institutions or citizens.

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More recently, we have witnessed several nations' "go open" campaigns, aimed at developing platforms whose architecture and resources are open and free to users. The concepts of "open data," "open educational resources," and "open learning" have been promoted as promising means for educational institutions to democratize data access and to share online resources. Governments in the United States, Europe, and Australia have actively invested in such platforms. In 2016, The US Department of Education, in a joint effort with the Department of Defense, started The Learning Registry. According to the website, The Learning Registry "acts as an aggregator of metadata, including the publisher, location, content area, standards alignment, ratings, reviews, and more" (Learning Registry 2016). Teachers and educators across the country can participate in the registry by submitting learning materials and metadata, where they will be vetted and made available for educational purposes. In the same year, the European Union commissioned its Open Education Europe Project (2017), mirroring the American government's effort to support the development of open educational resources and learning analytics systems. In contrast to the American project, the European Union wants to go beyond MOOCs to embrace openness along dimensions of technology and pedagogy but also to guarantee wider accessibility and collaboration between teachers (dos Santos, Punie, and Muñoz 2016).

Open access resources, in the form of data and content repositories and shared analytics and learning systems, are increasingly positioned as the "common good" rejoinder to proprietary data hubs and corporate learning platforms. Their potential to provide reservoirs and tools is supposed to save costs and stimulate teachers; furthermore, open resources are considered a boon to researchers and entrepreneurs who want to develop new learning materials and products. The many advantages of open source education are eagerly proclaimed; meanwhile, there have been relatively few concerns about potential hindrances or adverse effects in the long run, even if such reserve is probably warranted. Researchers favoring the use of open resources in education call for a fuller exploration of the effects of "open" education, addressing potential drawbacks and obstacles, such as limited sustainability, high costs, and debatable democratizing effects.

Just as in the corporate domain of educational platforms, the effects of online learning methods powered by open source tools have been scarcely researched. A few small-scale studies of open educational data projects in Australian schools were designed to empirically prove the effects of open source learning. Selwyn, Henderson, and Chao (2016) report the results of a collaborative project in two schools, where teachers and administrators deployed datafied systems and made teaching materials open and reusable. Testing the main claims of open resources and open data (i.e., enhanced engagement of teachers and students, more empowerment and accountability), the study could not confirm these outcomes. Instead, it showed that open data use is quite costly and that meaningful exploitation of data-based systems cannot be sustained in the long run without permanent extra funding. The democratic potential of open data projects in terms of enhanced accessibility and equality, as these researchers conclude, may be overstated. This conclusion appears to be corroborated by another Australian study of an open data-based public platform, MySchool, that resulted in a mixed bag of positive and negative effects (Marjanovic and Cecez-Kecmanovic 2017).²⁵

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In other words, "opening up" data and rendering platforms open does not in and of itself expedite democratic goals. Open data can empower teachers, students, and parents; but by the same token, it can deepen power imbalances and become a tool of disciplinary power. Collecting open source educational material and making it available on an open site may benefit teachers and entrepreneurs alike, but it takes appropriate tools and contextual materials to make it ready for (re)use. Making data, tools, and materials "open" is expensive; and appropriate funding is often what public school systems lack. As Rob Kitchin (2014) rightly points out, "no established model has been devised to sustainably fund open data initiatives beyond government subvention, and the benefits of open data in terms of producing additional consumer surplus value and new commercial ventures, innovative products and costs being offset by additional tax revenue are largely hypothetical or only beginning to materialise" (66). "Open" is not the equivalent of "public" in education; moreover, "open" does not mean "free." And finally, "open" is not a panacea for underfunded public education. If government support for open educational resources will siphon more public funding away from teaching in schools and doing research

at universities, it may actually be counterproductive. In other words, “open” infrastructures require a substantial commitment to the goals of public education, which brings us back to the clashing definitions of education as a common good.

Conclusion

The platformization of education has led to a fierce contestation of public values, which are traditionally anchored in institutions and professional codes that are increasingly bypassed and uprooted. The mechanisms of datafication, personalization, and commodification have penetrated deeply into the edifice of education, not only transforming the content of learning materials and students’ learning processes but also affecting pedagogical principles as well as the organization of schools and universities. Datafication and personalization indeed raise many social, ethical, and normative questions concerning the kind of education children may engage with in the future. As a result of commodification, learning data have become a valuable currency. In a connective world, technological interoperability is a vehicle for commercialization and educational platforms have become a battleground for conflicting private, corporate, and public interests. So the protection of education as a common good warrants extra vigilance in what is still considered a public sector, particularly in Europe. There are at least three concerns raised in this chapter that we will take to the final chapter’s reflections.

p. 135 First, the platform society is entangled in the ecosystem’s techno-commercial mechanisms that push the marketization of public services and public sectors. Since there is no public space carved out in the infrastructural core of platform society, technocratic solutions to social problems increasingly define the agendas of governments and public institutions. At the heart of this ideological clash lies not only what Morozov (2013) has called “technological solutionism” but also a neoliberal political agenda where formerly defined public and government functions are administered toward yielding private profits. Public funding is increasingly lured toward platform projects’ capitalizing on data-based, technology-intensive forms of teaching and learning, at the expense of investments in human-based, labor-intensive pedagogical and didactic skills. Datafication and personalization are pushed as the mantras of a new educational paradigm where human judgment is increasingly replaced by a product of predictive analytics that has acquired an aura of objectivity and empirical groundedness. However, there is as yet very little research evidence to substantiate the broad claims implied in this paradigm.

Second, the expansive ambitions of educational platforms as illustrated by their investment in scalable technology raises the question of efficacy: does scalability make education more effective or efficient? The automation of resources and learning systems raises the prospect of globalized “one-size-fits-all” education, a prospect that takes little account of the local and unique contexts of learning and teaching. Although platforms like AltSchool and Coursera promote “personalized mass education” as a unique selling point, what is behind the drive to scale technology is indeed global standardization. If online courses become the standard container for global education, shipping can be facilitated worldwide, from Shanghai to San Francisco, by a few global “connector” companies that create path-dependent distribution chains. While Coursera and Uber have much in common in terms of their underlying platform mechanisms, education differs from transportation in that the former is mostly about the uniqueness of each person, each place, and each cultural context.

Finally, the most profound insight taken from this chapter is how education, as traditionally part of the public sector that is uniquely entrusted with democratic public values, is rapidly inflected by the techno-commercial architecture of corporate platforms. As we have touched on in this chapter, there are notable attempts by governments, nonprofits, and civic groups to design a constellation of “open” platforms, aimed at helping teachers, students, and researchers work with data and digital tools. However, making data and knowledge *open* as we have argued is not a warranty to keep education *public*. Many projects for open educational

resources focus on the quantity and quality of data sets collected. What is needed besides affordable, manageable, and easily maintainable resources is a better and deeper understanding of how data flows reflect, construct, and enact *public values*. The same “open” data sets can be used toward different goals. For instance, student performance track records, depending on their use, may lead to better personalized attention by teachers but may also enhance discrimination or limit accessibility.

A proactive engagement with potential short-term and long-term effects of datafication and personalization should be accompanied by the drive to ask fundamental normative questions, not just about privacy, data security and surveillance but also about pedagogical foundations, the intrusiveness of techno-managerial surveillance systems, and the nature of quality education. Only if these wider societal norms and values are made transparent can we weigh and judge their implementation into platforms to which we trust our students’ education. We will take these three concerns and insights to the last chapter, where we will elaborate on the role of governments and civil society actors as actors in the platform society.

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Notes

1. In the Spring of 2017, Google’s influence in the classroom led to a poignant series of articles in the *New York Times*, reporting how entire school systems are infiltrated by high-tech companies, particularly the ones that are underfunded and in need of new equipment (Singer 2017a, 2017b).
2. The market for educational hardware and software in the United States has grown exponentially in the past decades. In 2015, US nursery schools and high schools spent more than \$13 billion on hardware (tablet, laptop, and desktop computers) and software (learning tools, administrative systems, digital content, etc.) (Singer, 2015). The American school computer-and-software market is expected to reach \$21 billion in sales by 2020 (Singer and Ivory 2017).
3. Amazon Inspire looks conspicuously similar to Amazon’s familiar shopping interface with features such as a search bar at the top of the page, user reviews, and star ratings for each product (Amazon Inspire 2017).
4. Most notably, the Chan-Zuckerberg Foundation has substantially invested in Summit Schools; Reed Hastings, chief executive officer of Netflix, has donated large sums to a nonprofit charter-school fund so that it could acquire DreamBox, a Silicon Valley educational program combining video games with math assignments. [Code.org](http://code.org) is a nonprofit consortium financed by Silicon Valley investors, which aims at teaching computer science in every public school in the United States. The latter example is typical of a skills-oriented initiative in online education (Singer 2017b). And in 2017, tech company Oracle announced an existing charter school, Design Tech High School, to be built on its campus in Redwood Shores in Silicon Valley.
5. Independent platforms in the educational market are, for instance, Tes.com, a site based in London with over eight million users worldwide; according to its mission statement, “educators can discover, share and sell original teaching materials” and through TES-Teach, a lesson-building product, “those resources can be freely integrated and implemented” through Wikispaces, an “open classroom-management platform that facilitates student-teacher communication and collaboration” (TES.com 2016). Another example is TeachersPayTeachers (TPT), a nonprofit initiative that later turned into a privately owned platform based in New York, “where teachers buy and sell original educational materials”; TPT engages over two million teachers in exchanging course materials on a variety of subjects (Teacherspayteachers 2016).
6. The center of educational governance, as Williamson (2016a) argues, “is being distributed and displaced to new digitized ‘centres of calculation’” (123). Massive information transfer to “calculation centres” and data hubs tallies with the concentration of power by a few large data companies, which may serve commercial rather than pedagogical interests (Lawson, Sanders, and Smith 2015).
7. As Mead (2016) explains in a profile in the *New Yorker*, AltSchool also opened a physical location in Brooklyn Heights; as of March 2018, AltSchool has two locations in San Francisco and two more in Brooklyn.
8. Reputational ranking and recommendation systems in educational platforms have been particularly criticized because they are informed by the techno-commercial logic of platforms; assessments happen *instantly* and *continuously*, mostly on the basis of *perception* or *likability* (Coetzee et al. 2014). However, educating is a process very different from “liking” or “recommending.” Students do not always like what they learn; learning often requires endless practice or involves unexpected encounters with content that only much later turn out to be valuable. The instantaneity of recommendations and likability of perceptions may be squarely at odds with long-term pedagogical values of curriculum-based education.
9. Pedagogical principles and teaching expertise are readily traded for managerial systems and information technology (IT) experts. As Hartong (2016, 530–31) observes: “In the perfect world of digital-era governance, state-organized educational

- institutions (such as schools) become gradually substituted with intelligent education networks, which operate as interactive online learning cultures, while schools and teachers are expected to secure IT-handling skills.”
10. In September 2015, Facebook’s Mark Zuckerberg announced he would start to fund the Summit Public Schools in San Francisco—an initiative that develops software to help create tailored lessons and projects, administer individualized quizzes that special software can grade, and track individual children to learn at their own pace. According to Chris Cox, Facebook’s chief product officer, the initiative “operates completely separate from Facebook and doesn’t require a Facebook account” while everybody working on the project “is subject to strict privacy controls that help protect student data” (Cox 2015).
 11. Although Coursera (backed by Stanford and private equity funds) is a for-profit platform and edX (backed by MIT and Harvard University) is a nonprofit initiative, it is not a clear-cut distinction; after all, MIT and Harvard are private universities with large financial resources. While Coursera and edX have different business models, the way they are driven by platform mechanisms and how they control data flows are very similar. For more information on business models in MOOCs, see Dellarocas and Van Alstyne (2013).
 12. Udacity, for instance, started in 2012 as a general MOOC for university students but has since developed into a platform aimed at professional education, mostly in the field of information and computer science.
 13. Initially backed by four American universities, Coursera, in 2017 engaged with 149 partnerships across 29 countries in North America, Europe, South America, and Asia. In 2017, Coursera had over 24 million registered users and offered more than 2,000 online courses.
 14. The estimated average production costs of a MOOC are \$50,000 per online course (including teachers’ salaries).
 15. Indeed, over the past ten years, there has been sharp criticism of the way in which colleges, particularly in the US, increasingly invest in material facilities at the expense of teaching and research. Universities spending too much money on facilities (sports facilities, buildings, etc.), tend to increase tuition and fees, rendering a college education decreasingly affordable to many students. In Europe, publicly funded university systems are still the norm, but some countries (e.g., the United Kingdom) have recently substantially raised tuition and fees for individual students.
 16. According to Bogen (2015), Coursera’s contracts with universities promise 6–15% in revenue-sharing per course with the university and 20% of gross profits on aggregate sets of courses, paid out every quarter.
 17. Another variant is the full master-degree program online, offered by a platform. In 2015, Coursera started to offer a degree MA program in collaboration with Georgia Tech University in exchange for an unknown percentage of tuition fees. As of 2017, Coursera offers several full Masters degrees, including several degree programs at the University of Illinois.
 18. As Coursera’s terms of service state, ‘If you participate in an online course, we may collect from you certain student-generated content, such as assignments you submit to instructors, peer-graded assignments and peer grading student feedback. We also collect course data, such as student responses to in-video quizzes, standalone quizzes, exams and surveys” (Coursera 2017).
 19. In response to deep cuts in state spending on higher education, American public colleges have enrolled fewer poor and middle-class students (Leonhardt 2017).
 20. The first studies on the efficacy of MOOCs and their ability to open up education, particularly a study by Hansen and Reich (2015), show that MOOC users on average stem from economically advantaged groups, so they increase rather than decrease inequality.
 21. The edX platform is backed by a large consortium of private and public universities, colleges and polytechnics (led by MIT and Harvard), nonprofit institutions, national governments, nongovernmental organizations, and multinational corporations, including Microsoft (edX 2017).
 22. There are some very successful examples of public infrastructural facilities that have enormously benefitted the sector of higher education; think of Eduroam, an international roaming service that allows teachers and students easy and secure network access when visiting educational institutions around the world.
 23. Walthausen (2016) published an interesting (professional and journalistic) inquiry into the usefulness of a variety of open resources, both paid for and so-called open resources promoted by the Department of Education’s Go Open campaign. She concludes that neither online resource provides ready-made classroom material because each teacher needs to repurpose the material for local, personalized classroom use. In other words, the repurposing of online course material remains a labor-intensive effort, in spite of “free” reusable materials.
 24. OpenupEd was the first pan-European MOOC initiative. It was launched in April 2013 by the European Association of Distance Teaching Universities in collaboration with the European Commission (OpenupEd 2017).
 25. The MySchool portal was launched by the Australian Curriculum Assessment Report Authority in January 2010. The platform’s original purpose was to make all school performance data open in order to improve teaching quality and enhance student performance. Positive effects were the growth of employment due to new online opportunities for teachers, but the researchers found several unintended consequences of datafication and personalization that were not anticipated. The most eye-catching one was a significant increase of anxiety levels among students, parents, and teachers

who could now see all individual performance indicators and compare them to other schools and school systems. This resulted in adverse effects ranging from the withdrawal or exclusion of low-performing students from some schools to an increase of real-estate prices in areas around well-performing schools, reinforcing inequality between students from different neighborhoods.